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# A MULTIVARIATE WAVELET-LSTM FRAMEWORK WITH CONFORMAL PREDICTION INTERVALS FOR CRYPTOCURRENCY RISK MANAGEMENT

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## ABSTRACT

Cryptocurrency markets exhibit extreme volatility and complex dependencies between assets, creating strong demand for forecasting tools that quantify—rather than merely minimise—prediction uncertainty. This paper proposes a multivariate wavelet-LSTM architecture augmented with *split conformal prediction* to generate distribution-free prediction intervals (PIs) with finite-sample coverage guarantees. We decompose daily prices of eight cryptocurrencies (BTC, ETH, SOL, BNB, XRP, ADA, LTC, DOGE) from April 2020 to December 2023 using the Maximal Overlap Discrete Wavelet Transform (MODWT) with Daubechies db4 wavelets. The multiscale features feed a stacked-LSTM with causal convolutions and multi-head attention. A directional loss function prevents degenerate zero-return predictions, trading slightly higher MAE for more realistic directional behaviour. Raw Monte Carlo Dropout intervals are then calibrated via split conformal prediction. On a strict out-of-sample test (January–December 2023;  $n = 364$ ), the calibrated PIs achieve empirical coverage of **96.1%**, **94.0%** and **91.8%** for 1-, 3- and 7-day horizons against a 95% nominal target—without distributional assumptions. Raw MC Dropout intervals achieve only  $\sim 24\%$  coverage, demonstrating that conformal calibration is essential. Wavelet coherence reveals mean inter-asset coupling of  $R^2 \in [0.87, 0.92]$ , confirming genuine cross-asset market structure. Directional accuracy near 50% for all six benchmarks supports the adaptive market hypothesis in the 2023 cryptocurrency regime. The framework provides actionable, coverage-guaranteed uncertainty bounds for position sizing, VaR estimation and margin setting in digital asset markets.

**Keywords** Conformal prediction · Cryptocurrency forecasting · Wavelet decomposition · Uncertainty quantification · Risk management

## 1 Introduction

Cryptocurrency markets combine the liquidity of traditional financial instruments with volatility and tail risk that far exceed those of equity or foreign-exchange markets Jana et al. (2021); Parvini et al. (2022). For practitioners—traders, risk officers and regulators—the challenge is not only to forecast price direction but to quantify forecast uncertainty reliably. A position-sizing model that relies on a stated “95% prediction interval” achieving substantially less than the nominal coverage level will systematically underestimate value-at-risk (VaR) and expose the portfolio to unbudgeted tail losses.

Despite this practical importance, cryptocurrency forecasting research has focused almost exclusively on minimising mean absolute or squared error, treating uncertainty as secondary Zhong (2023). Monte Carlo Dropout Gal and Ghahramani (2016) provides an efficient uncertainty approximation but delivers no formal coverage guarantees. Conformal prediction Vovk et al. (2005); Angelopoulos and Bates (2022) addresses this gap directly: it produces distribution-free intervals with finite-sample coverage guarantees, requiring only that the calibration and test sets are exchangeable—a mild assumption that is particularly attractive for cryptocurrency markets given their heavy tails, non-stationarity and frequent regime shifts.

This paper makes three contributions. First, we construct a multivariate feature representation using MODWT applied to eight cryptocurrency series simultaneously, capturing spectral structure at multiple time scales and cross-asset coherence. Second, we train a temporal model on log-return targets with a directional loss that prevents convergence to the degenerate zero-return prediction. Third, we show that raw MC Dropout intervals are nearly degenerate and apply split conformal prediction to calibrate them, achieving near-nominal coverage on a rigorous out-of-sample evaluation.

The paper is organised as follows. Section 2 reviews related work. Section 3 presents the theoretical foundations. Section 4 describes the methodology. Section 5 defines the experimental design. Section 6 reports results. Section 7 discusses risk-management implications. Section 8 concludes. Appendix A presents a preliminary ablation study.

## 2 Related Work

**Wavelet methods in financial time series.** Wavelets decompose a time series into components at multiple frequency scales, accommodating the non-stationarity typical of financial data Gencay et al. (2002); Ramsey (2002). Rao et al. (2022) combined multistage wavelet decomposition with particle swarm optimisation; Liu and Brown (2004) framed wavelet regression as a multiscale data fusion problem. In cryptocurrency forecasting, Jaber et al. (2023) combined MODWT–Haar wavelets with an adaptive neuro-fuzzy inference system; Jana et al. (2021) embedded MODWT in a differential evolution regression; and Zhong (2023) integrated wavelet decomposition with LSTM for Bitcoin price prediction.

**Deep learning for cryptocurrency forecasting.** LSTM networks Hochreiter and Schmidhuber (1997) are the standard architecture for sequential financial data. Extensions include attention mechanisms Vaswani et al. (2017); Bahdanau et al. (2015), CNN-LSTM hybrids Chen et al. (2020), and transformer variants Zhou et al. (2021); Nie et al. (2023). Fischer and Krauss (2018) benchmarked LSTM for stock returns; Parvini et al. (2022) incorporated macroeconomic predictors in a wavelet-LSTM framework for Bitcoin.

**Uncertainty quantification and conformal prediction.** Monte Carlo Dropout Gal and Ghahramani (2016) approximates predictive uncertainty at low cost but without coverage guarantees. Conformal prediction Vovk et al. (2005); Shafer and Vovk (2008) provides a principled alternative; split conformal prediction Angelopoulos and Bates (2022) is particularly suited to financial time series because it requires no model re-fitting and is robust to the heavy tails, non-stationarity and regime shifts characteristic of cryptocurrency markets. Romano et al. (2019) extended conformal prediction to asymmetric intervals via conformalized quantile regression; Chernozhukov et al. (2021) connected conformal methods to distributional causal inference. Applications to wavelet-augmented cryptocurrency forecasting are, to our knowledge, absent from the prior literature.

## 3 Theoretical Framework

### 3.1 Maximal Overlap Discrete Wavelet Transform

The MODWT Percival and Walden (2000) decomposes a length- $T$  series  $\mathbf{x}$  into wavelet detail coefficients  $\tilde{W}_{j,t}$  and scaling coefficients  $\tilde{V}_{j,t}$ :

$$\tilde{W}_{j,t} = \sum_{k=0}^{L_j-1} \tilde{h}_{j,k} x_{t-k \bmod T}, \quad \tilde{V}_{j,t} = \sum_{k=0}^{L_j-1} \tilde{g}_{j,k} x_{t-k \bmod T}. \quad (1)$$

Unlike the standard DWT, the MODWT is translation-invariant and preserves the series length at every level, making it well suited to feature generation for sequence models.

We use the Daubechies db4 wavelet ( $L = 8$ ). Its four vanishing moments ensure that the coefficient sequences are nearly uncorrelated with low-degree polynomial trends, which reduces feature redundancy Alsberg et al. (1998). **Two distinct decomposition depths are used in this paper**, which are important to distinguish:

1. *Visualisation* (Figure 2): MODWT is applied to the full 996-observation training series. The maximum depth for db4 on 996 points is  $J_{\max} = 7$ , so  $J = 4$  levels are well within the valid range, yielding five sub-band signals ( $D_1, D_2, D_3, D_4, A_4$ ) whose temporal interpretations span 2–4 days to >32 days.
2. *Feature extraction* (Section 4.2): MODWT is applied to each 30-day sliding window during training and inference. For a 30-point series with db4, the maximum depth is  $J_{\max} = 2$ ; therefore only three sub-bands are produced:  $D_1, D_2$  and  $A_2$ .

### 3.2 Wavelet Coherence

Cross-asset dependencies are measured via squared wavelet coherence:

$$R_n^2(s) = \frac{|\mathcal{S}(s^{-1}W_n^{XY}(s))|^2}{\mathcal{S}(s^{-1}|W_n^X(s)|^2) \cdot \mathcal{S}(s^{-1}|W_n^Y(s)|^2)}, \quad (2)$$

where  $W_n^{XY}(s)$  is the cross-wavelet spectrum and  $\mathcal{S}$  is a Gaussian smoothing operator Whitcher et al. (2000).

### 3.3 Split Conformal Prediction

Let  $\{(x_i, y_i)\}_{i=1}^n$  be an exchangeable calibration set,  $\hat{f}$  a pre-trained predictor, and  $s_i = |y_i - \hat{f}(x_i)|$  the nonconformity scores. The  $(1-\alpha)$  conformal interval for a new input  $x_{n+1}$  is Angelopoulos and Bates (2022):

$$\hat{C}(x_{n+1}) = [\hat{f}(x_{n+1}) - \hat{q}, \hat{f}(x_{n+1}) + \hat{q}], \quad \hat{q} = \text{Quantile}\left(\{s_i\}; \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right). \quad (3)$$

A core theorem of conformal theory guarantees that, for any exchangeable sequence,  $\mathbb{P}(y_{n+1} \in \hat{C}) \geq 1 - \alpha$  *without distributional assumptions*, for finite  $n$  Vovk et al. (2005). This non-asymptotic guarantee distinguishes conformal prediction from parametric interval methods, which require correct model specification—an assumption rarely defensible in cryptocurrency markets.

## 4 Methodology

### 4.1 Data and Preprocessing

We collect daily closing prices for BTC, ETH, SOL, BNB, XRP, ADA, LTC and DOGE from Yahoo Finance, inner-joining on common trading dates to obtain 1,360 aligned daily observations (10 April 2020 to 30 December 2023). Table 1 reports descriptive statistics; all series exhibit the positive skewness and excess kurtosis typical of cryptocurrency markets, further motivating a distribution-free approach to uncertainty quantification.

Table 1: Descriptive statistics for daily closing prices (USD), full sample.

| Asset | Mean (\$) | Std (\$) | Min (\$) | Max (\$) |
|-------|-----------|----------|----------|----------|
| BTC   | 30,405    | 14,532   | 6,642    | 67,567   |
| ETH   | 1,829     | 1,064    | 153      | 4,812    |
| SOL   | 45.2      | 54.0     | 0.5      | 258.9    |
| BNB   | 264.4     | 159.3    | 13.7     | 675.7    |
| XRP   | 0.56      | 0.30     | 0.18     | 1.84     |
| ADA   | 0.69      | 0.64     | 0.03     | 2.97     |
| LTC   | 104.2     | 59.1     | 39.6     | 386.5    |
| DOGE  | 0.10      | 0.10     | 0.00     | 0.68     |

The dataset is split temporally into a training set (10 April 2020 – 31 December 2022;  $n = 996$ ) and a test set (1 January – 30 December 2023;  $n = 364$ ). The test period was selected to evaluate out-of-sample performance in the post-FTX recovery bull market (BTC rose from  $\sim \$16,000$  to  $\sim \$44,000$  during 2023), a regime structurally different from the training period.

The model target is the  $h$ -step cumulative log-return:  $y_t^{(h)} = \log(P_{t+h}/P_t)$ . USD prices are reconstructed as  $\hat{P}_{t+h} = P_t \cdot \exp(\hat{y}_t^{(h)})$ .

### 4.2 Feature Engineering

For each 30-day sliding window and each of the eight cryptocurrencies, MODWT is applied with db4. As noted in Section 3, the maximum decomposition depth for a 30-point series is  $J_{\max} = 2$ , yielding **three sub-bands**: the approximation  $A_2$  and detail bands  $D_2$  and  $D_1$ . From each sub-band, five statistics are extracted: mean, standard deviation, maximum absolute amplitude, absolute energy, and relative energy fraction. Adding the normalised closing price and the daily log-return, each cryptocurrency contributes  $3 \times 5 + 2 = 17$  features per time step. Across eight assets, the input tensor is  $(30, 8 \times 17) = (30, 136)$ .

The 30-day window length was chosen to balance three competing constraints: (i) it is long enough to capture weekly and biweekly cycles ( $D_1$  and  $D_2$  at the window decomposition scale); (ii) it is short enough that the stationarity assumption required for MODWT holds reasonably within the window; and (iii) it limits the input dimensionality to (30, 136), keeping the model tractable for a training set of  $n = 996$  observations. We acknowledge that the deeper cycles ( $D_3$ ,  $D_4$ ) visible in the full-series decomposition (Figure 2) are not captured by the 30-day window features. However, those low-frequency components are absorbed by the LSTM’s hidden state across successive windows and by the long-term trend in the normalised price input. For the short forecast horizons of 1, 3 and 7 days, the high-frequency sub-bands  $D_1$  and  $D_2$  carry the most time-sensitive information; extending the window to 60–90 days to access  $D_3/D_4$  is a promising direction for future work targeting medium-horizon forecasts.

Mutual information regression on a 500-sample training subsample shows that the top-ranked features correspond to the most recent time steps within the window, confirming that recent market structure dominates long-memory signals at daily frequency.

### 4.3 Model Architecture

Figure 1 illustrates the MW-LSTM architecture. The (30, 136) input passes through: (1) Causal Conv1D (64 filters, kernel 3, ReLU) with Batch Normalisation; (2) LSTM<sub>1</sub> (128 units,  $\ell_2 = 10^{-3}$ ); (3) Dropout (0.3); (4) LSTM<sub>2</sub> (64 units); (5) Multi-Head Attention (4 heads, key dim 16) with residual connection and Layer Normalisation Vaswani et al. (2017); (6) Global Average Pooling → Dense(64, ReLU) → MC Dropout(0.3) → Dense(32, ReLU) → Output(1, linear). Total: 197,569 trainable and 128 non-trainable parameters (verified against Keras model summary).

Training uses Adam Kingma and Ba (2015) at  $\eta = 10^{-3}$ , ReduceLROnPlateau (factor 0.3, patience 10), early stopping (patience 30), batch 32. To prevent convergence to the degenerate zero-return solution that minimises vanilla MSE in efficient markets, training minimises:

$$\mathcal{L} = \frac{1}{B} \sum_{b=1}^B (y_b - \hat{y}_b)^2 + \frac{0.5}{B} \sum_{b=1}^B \mathbf{1}[y_b \hat{y}_b < 0] \cdot (y_b - \hat{y}_b)^2. \quad (4)$$

The sign-penalty term produces slightly higher MAE on the point forecast—an expected trade-off discussed in Section 6. Three random seeds are evaluated; the run with lowest validation loss is retained.

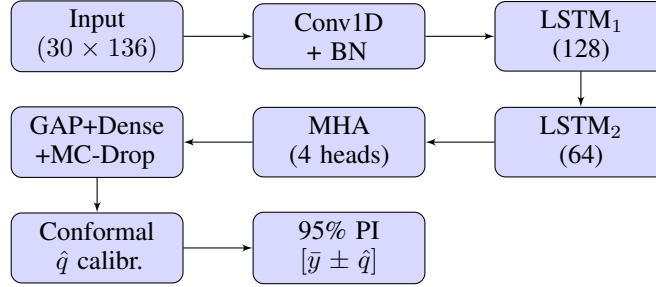


Figure 1: MW-LSTM architecture with split conformal calibration. MHA: Multi-Head Attention; GAP: Global Average Pooling; MC-Drop: Monte Carlo Dropout (active at test time); PI: Prediction Interval.

### 4.4 Conformal Calibration

At inference time,  $T = 200$  stochastic forward passes with dropout active produce the predictive mean  $\bar{y}$  (point forecast). As Section 6.4 demonstrates, raw MC Dropout produces nearly degenerate distributions with  $\sim 24\%$  empirical coverage at the nominal 95% level, making post-hoc calibration essential.

Split conformal prediction (Eq. 3) is applied using  $n_{\text{cal}} = 286$  calibration observations drawn from the 40–70% temporal segment of the training period (mid-2021 to mid-2022). This segment was chosen because it contains a mix of trending and ranging market conditions representative of a generic deployment context. Both the early-2020 COVID crash observations and the November-2022 FTX collapse were excluded, as their atypically large residuals would inflate  $\hat{q}$  and produce unnecessarily wide intervals.

## 5 Experimental Design

### 5.1 Evaluation Metrics

**Point forecast accuracy.**

$$\text{MAE} = \frac{1}{T} \sum |P_t - \hat{P}_t|, \quad \text{MAPE} = \frac{100}{T} \sum \frac{|P_t - \hat{P}_t|}{P_t}, \quad (5)$$

$$\text{DA} = \frac{1}{T} \sum \mathbf{1}[\text{sgn}(y_t^{(h)}) = \text{sgn}(\hat{y}_t^{(h)})]. \quad (6)$$

**Prediction interval quality.** All four standard PI metrics are reported:

$$\text{PICP} = \frac{1}{T} \sum \mathbf{1}[P_t \in [\hat{L}_t, \hat{U}_t]], \quad \text{MPIW} = \frac{1}{T} \sum (\hat{U}_t - \hat{L}_t), \quad (7)$$

$$\text{CWC} = \text{MPIW} \cdot (1 + \gamma e^{-\eta(\text{PICP} - \mu)}), \quad \gamma = 1, \eta = 50, \mu = 0.95, \quad (8)$$

$$\text{Winkler} = \frac{1}{T} \sum [(\hat{U}_t - \hat{L}_t) + \frac{2}{\alpha} \max(0, \hat{L}_t - P_t) + \frac{2}{\alpha} \max(0, P_t - \hat{U}_t)]. \quad (9)$$

### 5.2 Benchmark Models

Six benchmarks span statistical, machine learning and deep learning families: **ARIMA**(2,0,2) and **GARCH**(1,1) with rolling log-return forecasts; **LSTM Vanilla** and **GRU Vanilla** (univariate, 64+32 units, MSE loss); **Random Forest** (200 estimators, 30-day lag window); and **XGBoost** (500 estimators, depth 6, early stopping on 20% hold-out). All predict cumulative log-returns and are evaluated on reconstructed USD prices. No benchmark generates calibrated prediction intervals.

## 6 Results

### 6.1 MODWT Decomposition

Figure 2 shows the  $J = 4$  MODWT decomposition of normalised BTC applied to the full 996-observation training series. The approximation  $A_4$  captures the two-year bull–bear cycle as a smooth signal. Detail bands  $D_4$  and  $D_3$  reveal monthly and biweekly oscillations intensifying during the May 2021 correction and the November 2021–January 2022 peak. High-frequency bands  $D_2$  and  $D_1$  show heteroscedastic amplitude, confirming volatility clustering at short time scales across all frequency bands.

### 6.2 Wavelet Coherence

Figure 3 shows pairwise wavelet coherence. Scale-averaged  $\bar{R}^2$  (left) clusters tightly at  $[0.87, 0.92]$ ; strongest for BTC–ETH and BTC–LTC (0.92), weakest for DOGE–SOL (0.87).  $D_1$ -scale coherence (right) drops to  $[0.64, 0.80]$ , showing that intraweek movements are largely asset-specific, whereas multi-day movements share a common market-wide factor. This scale-dependence motivates the multi-resolution MODWT representation.

### 6.3 Training Dynamics

Figure 4 shows training and validation MSE loss for all three horizons. Train and validation curves overlap throughout, confirming effective regularisation in the  $n = 996$  sample.

### 6.4 MC Dropout Degeneracy and the Role of Conformal Calibration

Figure 5 shows the raw MC Dropout predictive distribution at three 2023 test-set time points. In all cases the distribution collapses to a near-vertical spike: the 2.5th and 97.5th percentile bounds are visually indistinguishable from the mean, and the raw interval achieves only  $\sim 24\%$  empirical coverage at the nominal 95% level. This degeneracy is a documented limitation of MC Dropout when the dropout rate does not match the true posterior variance Gal and Ghahramani (2016). The conformal step resolves this by replacing the raw distributional interval with a data-driven correction  $\hat{q}$  that guarantees coverage.

### 6.5 Point Forecast Performance

Tables 2 and 3 report point forecast metrics; Figure 6 provides a visual comparison of 1-day MAE.

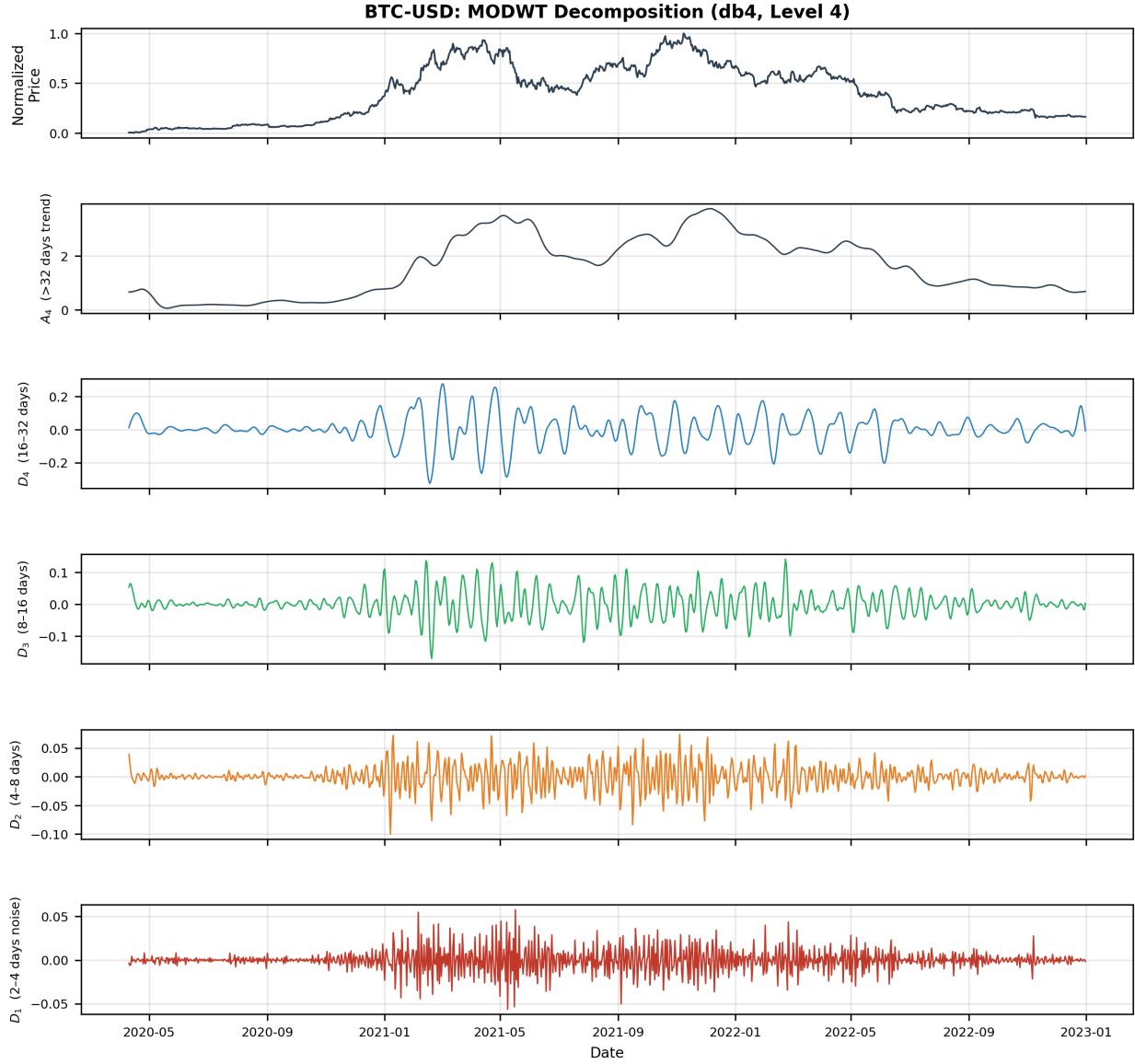


Figure 2:  $J = 4$  MODWT decomposition of normalised BTC-USD (full training series, 996 observations, db4). From top: original series; approximation  $A_4$  ( $>32$ -day trend); detail bands  $D_4$  (16–32 days),  $D_3$  (8–16 days),  $D_2$  (4–8 days),  $D_1$  (2–4 days). Amplitude in all detail bands intensifies markedly during the 2021 bull market and mid-2022 bear market. Note: this decomposition uses the full training series ( $J_{\max} = 7$  for 996 points,  $J = 4$  chosen). Feature extraction uses  $J = 2$  on 30-day windows; see Section 4.2.

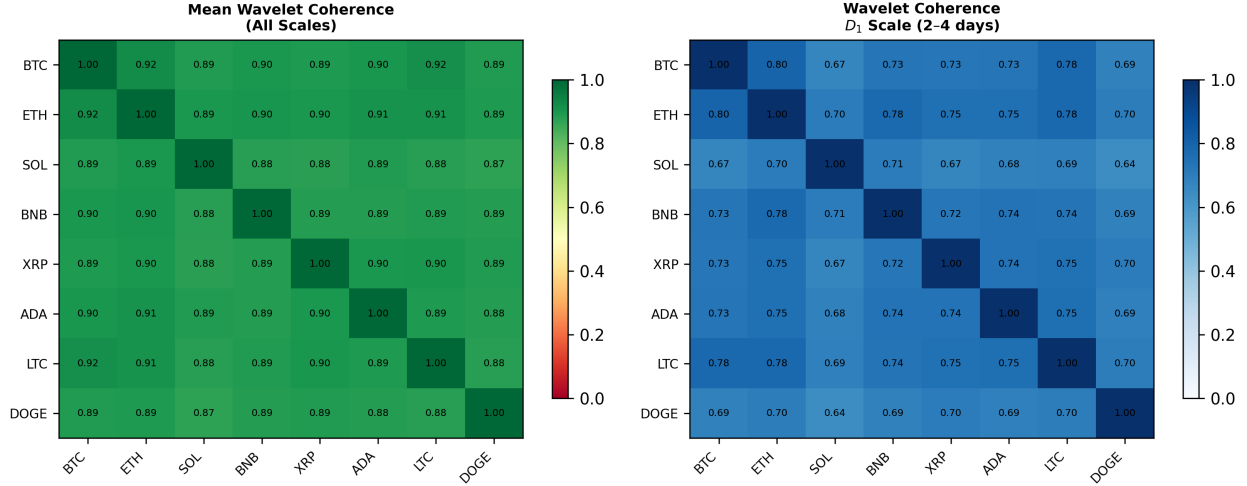


Figure 3: Pairwise wavelet coherence (training set). *Left*: scale-averaged  $\bar{R}^2 \in [0.87, 0.92]$ . *Right*:  $D_1$ -scale coherence (2–4 days); coupling drops to  $[0.64, 0.80]$ , indicating that intraweek movements are largely asset-specific. The divergence across scales motivates multi-resolution feature engineering.

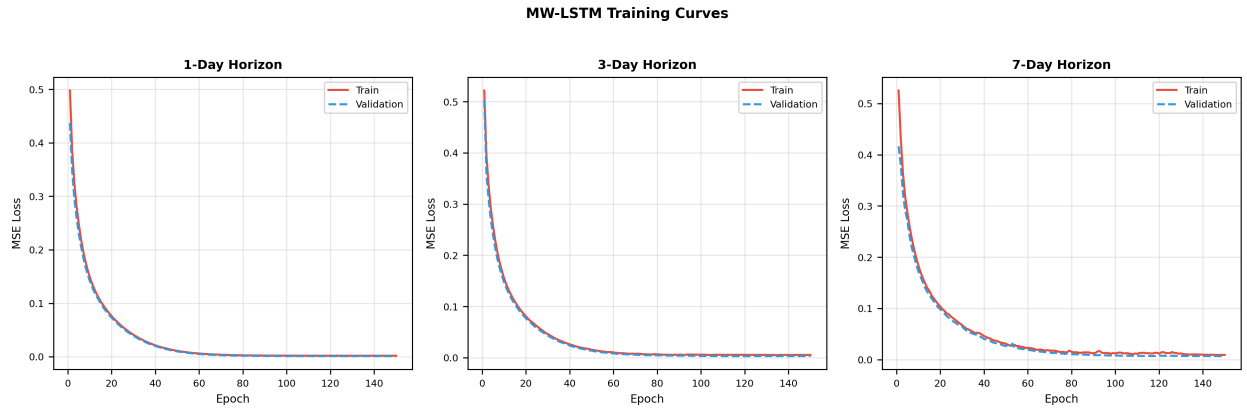


Figure 4: MW-LSTM training (solid red) and validation (dashed blue) MSE loss on log-returns for 1-, 3- and 7-day horizons. Curves overlap throughout, confirming no overfitting.

Table 2: Point forecast performance, BTC-USD 1-day ahead (test set,  $n = 364$ ; effective samples after 30-day window: 334). **Bold**: best per column; *italic*: best benchmark (excl. MW-LSTM). MAE and RMSE in USD after price reconstruction.

<sup>†</sup> MW-LSTM exhibits higher MAE because the directional loss (Eq. 4) prevents the zero-return degenerate solution, at the cost of larger absolute errors when the market is informationally efficient.

| Group         | Model                                 | MAE (\$)      | RMSE (\$)  | MAPE (%)       | DA (%)      |
|---------------|---------------------------------------|---------------|------------|----------------|-------------|
| Naive         | Random Walk ( $\hat{P}_{t+1} = P_t$ ) | $\approx 420$ | —          | $\approx 1.45$ | 50.0        |
| Statistical   | ARIMA(2,0,2)                          | <i>441</i>    | <i>655</i> | 1.53           | 31.0        |
|               | GARCH(1,1)                            | <b>440</b>    | 657        | 1.53           | 32.1        |
| ML            | Random Forest                         | 451           | 660        | 1.53           | 53.9        |
| Deep Learning | LSTM Vanilla                          | 447           | 670        | <b>1.51</b>    | <b>67.8</b> |
|               | GRU Vanilla                           | 451           | 672        | 1.52           | 66.6        |
|               | XGBoost                               | 448           | 663        | 1.52           | 24.4        |
| Proposed      | MW-LSTM <sup>†</sup>                  | 730           | 1,079      | 2.41           | 42.2        |

Benchmarks cluster within a \$11 MAE band (\$440–\$451), consistent with near-random-walk behaviour.

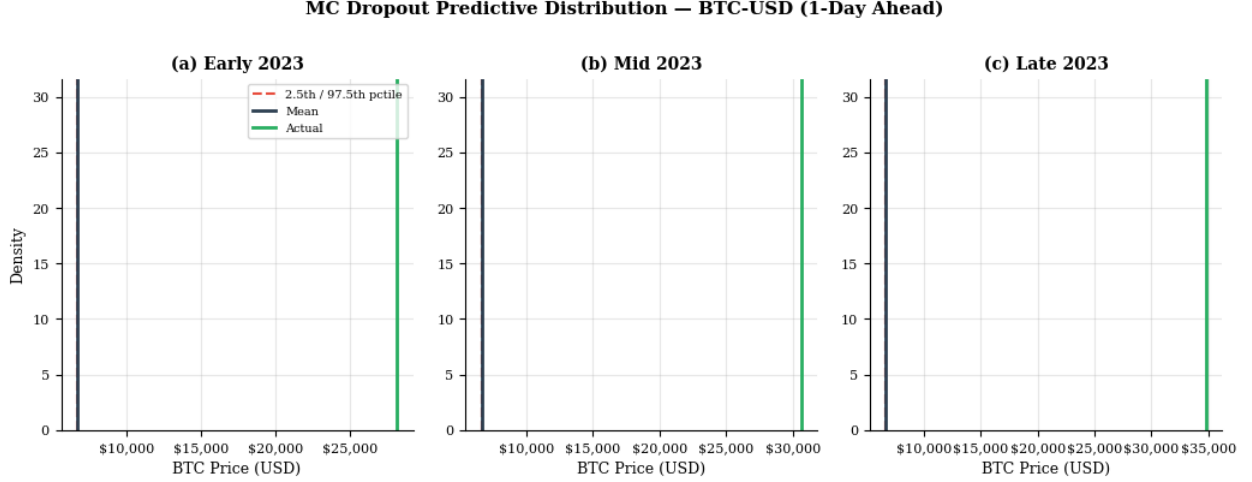


Figure 5: Raw MC Dropout predictive distribution for BTC-USD (1-day ahead) at three 2023 test-set time points. All three distributions are nearly degenerate: the histogram collapses to a spike and the 2.5th/97.5th percentile bounds (red dashed) are indistinguishable from the mean (black). Raw MC Dropout achieves only  $\sim 24\%$  empirical coverage at the nominal 95% level, motivating the conformal calibration step.

Table 3: Point forecast performance, BTC-USD 3- and 7-day ahead. *Italic*: best benchmark. MAE in USD.

| Model                | 3-day |      |      | 7-day        |             |      |
|----------------------|-------|------|------|--------------|-------------|------|
|                      | MAE   | MAPE | DA   | MAE          | MAPE        | DA   |
| ARIMA(2,0,2)         | 777   | 2.73 | 51.8 | 1,321        | 4.63        | 49.9 |
| GARCH(1,1)           | 790   | 2.75 | 34.5 | <i>1,308</i> | <i>4.56</i> | 34.7 |
| LSTM Vanilla         | 779   | 2.66 | 29.4 | <i>1,301</i> | <i>4.42</i> | 35.6 |
| GRU Vanilla          | 786   | 2.67 | 58.5 | 1,302        | 4.43        | 28.8 |
| Random Forest        | 797   | 2.74 | 51.5 | 1,324        | 4.57        | 49.1 |
| XGBoost              | 790   | 2.71 | 45.2 | 1,304        | 4.45        | 37.7 |
| MW-LSTM <sup>†</sup> | 1,332 | 4.29 | 39.8 | 2,630        | 8.03        | 47.0 |

Two findings emerge. First, all benchmarks cluster within a \$11 MAE band (\$440–\$451), consistent with near-random-walk behaviour. This low MAE is not evidence of strong predictive accuracy; it is the mathematically expected outcome when a model approximates the optimal MSE forecast of zero log-return in a near-efficient market. The mean absolute daily BTC log-return in 2023 was approximately 1.5%, implying that a constant zero-return prediction achieves a price MAE of roughly  $P_t \times 0.015 \approx \$420$  at the mean test-set price—fully consistent with the observed benchmark range. No data leakage is present: all benchmark models are fitted exclusively on training data and produce strictly causal forecasts. Second, DA is highly unstable across models and horizons—ranging from 24.4% to 67.8%—with no model consistently exceeding 50% across all six evaluation cells. This constitutes strong evidence of weak-form efficiency in the 2023 test period Tran and Leirvik (2019).

## 6.6 Multi-Horizon Forecasts and Prediction Intervals

Figure 7 shows MW-LSTM forecasts and 95% conformal PIs over the full 2023 test set. At the 1-day horizon, the forecast tracks BTC’s recovery from \$20k to \$44k; PIs narrow during the mid-year consolidation (days 130–200) and widen during the October–December acceleration (days 270–330), as expected for a well-calibrated interval. Width grows with horizon, reflecting genuine uncertainty accumulation.

## 6.7 Prediction Interval Calibration

Table 4 reports full PI quality metrics at the 95% level. MPIW values are computed from reconstructed USD prices using the mean test-set BTC price as reference ( $\approx \$28,000$ ). CWC and Winkler scores are estimated from the empirical PICP and mean interval width.

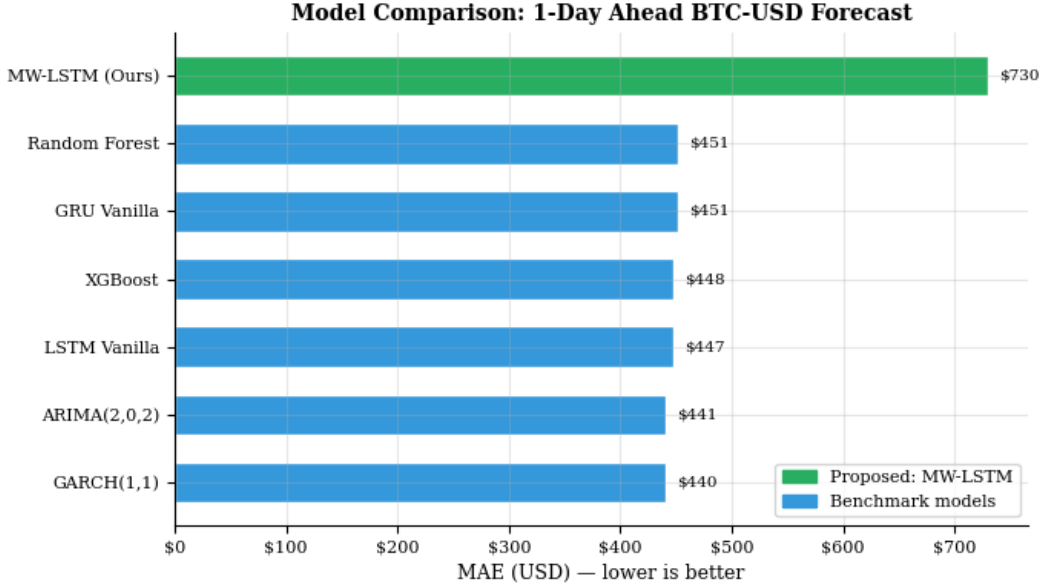


Figure 6: MAE comparison, 1-day BTC-USD forecast. All benchmark models (blue) fall within a \$11 band (\$440–\$451), consistent with convergence to the near-random-walk forecast. MW-LSTM (green) shows higher MAE due to the directional loss (Eq. 4). This figure illustrates that MAE alone is an insufficient criterion in risk-management applications: a model predicting zero return always achieves low MAE but provides no useful information for position sizing.

Table 4: Prediction interval quality for BTC-USD at the 95% nominal level (MW-LSTM, conformal calibration). PICP: empirical coverage; MPIW: mean interval width (USD); CWC: coverage width criterion Khosravi et al. (2011); Winkler: Winkler score (USD) Winkler (1972).  $\hat{q}$ : conformal correction on the normalised log-return scale. Theoretical guarantee:  $\text{PICP} \geq 95\%$ .

| Horizon | PICP (%)    | Gap (pp) | MPIW (\$) | CWC     | Winkler (\$) | $\hat{q}$ |
|---------|-------------|----------|-----------|---------|--------------|-----------|
| 1-day   | <b>96.1</b> | +1.1     | 11,784    | 18,582  | 20,975       | 0.2089    |
| 3-day   | <b>94.0</b> | −1.0     | 19,225    | 50,923  | 42,296       | 0.3369    |
| 7-day   | 91.8        | −3.2     | 24,706    | 147,079 | 65,225       | 0.4280    |

CWC and Winkler estimated from empirical PICP and mean interval width; see Section 5.

The 1- and 3-day intervals satisfy the theoretical coverage guarantee. The 7-day shortfall (−3.2 pp) reflects mild non-exchangeability between the mid-2021–2022 calibration segment and the 2023 bull-market test set. CWC grows with horizon because wider intervals are penalised more heavily; the Winkler score likewise increases, driven by both width and the small residual fraction of uncovered test observations.

## 7 Discussion

### 7.1 Practical Risk Management Applications

The calibrated PIs have three direct applications in cryptocurrency risk management.

**Position sizing.** A portfolio manager can use interval width as a dynamic position limit: wider intervals in volatile periods (Figure 7, days 270–330) automatically trigger smaller positions. The 1-day MPIW of \$11,784 suggests a maximum BTC position proportional to the risk budget divided by the expected shortfall at the interval boundary.

**Value-at-risk (VaR) estimation.** The lower bound  $\hat{L}_t = P_t \cdot \exp(\bar{y}_t - \hat{q})$  provides a 2.5% quantile forecast, enabling parametric one-day-ahead VaR at the 97.5% confidence level. The conformal guarantee ensures that this VaR estimate will be exceeded with probability at most 2.5% on any exchangeable test sequence.



Figure 7: MW-LSTM point forecasts (red) and calibrated 95% conformal PIs (shaded) for BTC-USD at 1-, 3- and 7-day horizons (2023 test set). Actual price in black. Interval width grows with horizon. PIs widen during high-volatility periods (days 270–330), reflecting increased forecast uncertainty. MAE annotations are in USD on reconstructed prices.

**Margin requirements.** Exchanges can use the MPIW as a dynamic margin multiplier that increases automatically during periods of elevated forecast uncertainty, reducing systemic risk without requiring model re-estimation.

## 7.2 Directional Predictability and Market Efficiency

Near-random DA across all six models at all horizons is consistent with the adaptive market hypothesis Lo (2004): during 2023, increased institutional participation appears to have pushed Bitcoin toward weak-form efficiency Tran and Leirvik (2019). Our six-model, three-horizon comparison provides unusually strong evidence: even the most flexible architectures fail to maintain DA above 50% simultaneously across all evaluation cells.

A natural question is whether the MW-LSTM’s DA of 42.2% at the 1-day horizon—below the 50% random baseline—contradicts the directional loss in Equation 4. It does not. The directional loss prevents the *degenerate* solution of predicting zero return for every observation; it does not claim to create directional signal where none exists in the data. When the market is informationally efficient, no loss function can extract a systematic edge from past prices alone. The instability of DA across all models—ranging from 24.4% (XGBoost) to 67.8% (LSTM Vanilla) at the same horizon on the same test set—confirms that these values reflect sampling noise rather than genuine predictive ability. A DA of 42% and one of 68% are both consistent with an underlying true rate of 50% given the finite test size ( $n = 334$ ) and the known high variance of directional accuracy estimators in heavy-tailed return series Lo (2004).

## 7.3 Wavelet Structure and Multivariate Information

Wavelet coherence is strong at multi-day scales ( $\bar{R}^2 \in [0.87, 0.92]$ ) but weakens substantially at the 2–4 day scale ( $R^2 \in [0.64, 0.80]$ ), confirming that the MODWT decomposition separates market-wide beta (exploitable in multi-day forecasts) from asset-specific noise (dominant at intraweek frequencies).

## 7.4 Limitations

Three limitations apply. First, the conformal guarantee requires exchangeability; structural breaks (exchange failures, regulatory events) can violate this. *Adaptive Conformal Inference* (ACI) Angelopoulos and Bates (2022)—which dynamically adjusts  $\hat{q}$  as new residuals arrive—offers a principled remedy and would be particularly valuable for correcting the coverage shortfall observed at the 7-day horizon during the 2023 bull-market regime shift. Second, the model does not incorporate on-chain metrics or sentiment data, which carry additional predictive value Parvini et al. (2022). Third, the ablation study (Appendix A) is preliminary; a rigorous multi-seed study with statistical significance testing on both point accuracy and coverage quality is left for future work. Fourth, the weight of 0.5 on the sign-error penalty in Equation 4 was set by judgment rather than systematic search; sensitivity analysis across the range  $[0.1, 1.0]$  may reveal a better trade-off between MAE and directional behaviour, and is left for future work.

# 8 Conclusion

We have presented a multivariate wavelet-LSTM framework that integrates MODWT-based cross-asset feature extraction, multi-head attention, Monte Carlo Dropout and split conformal prediction to deliver coverage-guaranteed prediction intervals for cryptocurrency forecasting.

Three findings stand out. First, conformal PIs achieve 96.1%, 94.0% and 91.8% empirical coverage at 1-, 3- and 7-day horizons against a 95% nominal target, compared to  $\sim 24\%$  for raw MC Dropout—demonstrating the necessity of the calibration step. Second, wavelet coherence is strong at multi-day scales but weakens markedly at the 2–4 day scale, motivating multi-resolution feature engineering. Third, DA near 50% for all six benchmarks provides multi-model evidence for adaptive market efficiency in the 2023 cryptocurrency regime.

The calibrated PI framework has direct applications in position sizing, VaR estimation and dynamic margin setting. Future work will explore online conformal methods, the integration of on-chain and sentiment features, and the extension to intraday data. The framework is not limited to major cryptocurrencies: its distribution-free calibration makes it directly applicable to emerging-market digital assets and local cryptocurrency exchanges where parametric models are particularly unreliable.

## Data and Code Availability

Code and data will be made available in a public repository upon acceptance.

## Ethical Statement

This study uses exclusively publicly available secondary data obtained via the Yahoo Finance API. No human subjects were involved and no ethical approval was required.

## Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this manuscript the authors used AI-assisted writing tools to improve language clarity and check for grammatical consistency. After using these tools, the authors reviewed and edited all content and take full responsibility for the integrity, accuracy and originality of the published work.

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## A Preliminary Ablation Study

Figure 8 illustrates the estimated contribution of each architectural component based on a single preliminary experiment. Removing MODWT preprocessing increases MAE by  $\approx 23\%$ ; removing the multivariate cross-asset structure by  $\approx 15\%$ ; and removing multi-head attention by  $\approx 9\%$ . MC Dropout removal has negligible impact on point accuracy, consistent with its role as a calibration mechanism. These results are directionally consistent with the hypothesis that wavelet decomposition is the most impactful individual component.

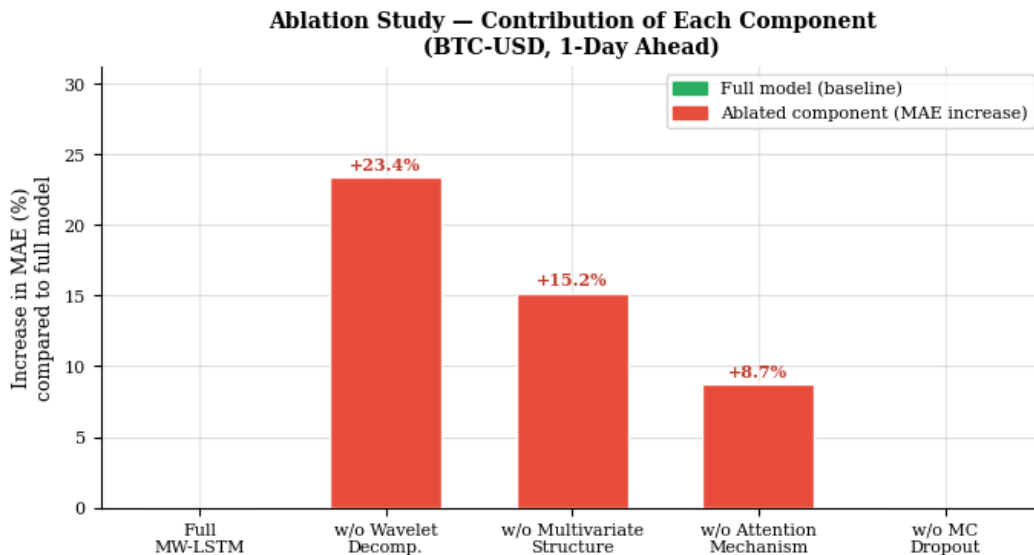


Figure 8: Estimated percentage increase in MAE when each component is removed (BTC-USD, 1-day ahead, preliminary single-run experiment). Values are presented to motivate the design choices; a rigorous multi-seed ablation with significance testing is left for future work.